

CSCI 335-02 Syllabus

Hunter College, Summer 2026

Instructor: Sadab Hafiz

<https://hunter-college-cs335-summer-2026.github.io/course-webpage/>

Mon|Tue|Wed|Thu 09:50 am - 11:24 am ET, Hunter West 207

sh3646@hunter.cuny.edu

1 Prerequisites

A student **must** complete CSCI 23500 and MATH 15500 with a C or better. This course builds upon the things taught in CSCI 23500. Entering this course, you are assumed to be proficient with linear data structures (including lists, stacks, and queues) and should be able to implement them as vectors and/or arrays in C++. You should also be able to write and use classes in C++, write well-organized code in accordance with OOP principles. Thus, you will have a very difficult time in this class without fulfilling the prerequisites.

2 Course Objectives

We'll study the design, analysis, implementation, and uses of a variety of data structures and algorithms. Data structures we'll cover this semester include self-balancing trees, hash maps, heaps, sets, and graphs. We will focus on algorithmic analysis and use of C++ STL to efficiently solve programming problems. All code should be as efficient and memory-safe as reasonably possible in C++, and should adhere to professional C++ best practices. While C++ is a key part of this course, the goal of this course is to provide an understanding of the concepts so that they can be applied in any programming language you learn in the future.

3 Learning Goals

Students who successfully complete this course will be able to do the following:

- Demonstrate the basic operations of the data structures we cover.
- Prove the computational complexities of basic operations of the data structures we cover.
- Relate the data structures we study to C++ STL containers.
- Write professional, efficient, memory-safe code using the C++ STL.
- Analyze the computational complexity of algorithms and code samples.
- Design and implement efficient solutions to programming problems.
- Explain trade-offs between different algorithmic solutions.

4 Textbook and Tools

The use of textbooks for this course is **optional**. If you plan to use a textbook, the recommended textbook is *Data Structures and Algorithm Analysis in C++*, 4th Edition, Mark Allen Weiss, ISBN-13: 978-0-13-284737-7.

We will mainly use C++ programming language. This course is taught on Linux and your programs must be able to run on a Linux environment. The standard Linux/Unix/MacOS C++ compiler is g++. Macs have Unix command line and g++. If you want a Linux environment on Windows without installing Linux, follow this [excellent tutorial](#) created by Moody. You can also install a C++ compiler on Windows using [MinGW](#).

The use of an IDE is recommended. If you prefer to code in your terminal directly, take a look at [Vim](#). If you prefer GUI, I recommend [VS Code](#) for its simplicity and flexibility. I will use VS Code to show examples during class. However, feel free to explore different IDEs and choose whichever one works best for you.

5 Office-Hours and Communication

Office hours will generally be after class. The format and frequency of the office hours will be announced and updated in the course webpage. Important announcements will be made via Brightspace and in class. I recommend checking your student email frequently as updates in Brightspace are sent to the linked email.

6 Course Materials

All the materials covered will be available on [course webpage](#). Make sure to stay up-to-date with new materials, and assignments deadlines. We will use GitHub and Gradescope for assignments.

7 Grade Distribution

You can expect to receive your grades in a timely manner. Your final grade will be based on the following:

Final Exam	30%
Midterm Exam	15%
Projects	45%
Weekly Quizzes	10%

There will be no negotiation about the grade distribution. The number of quizzes and projects may change depending on the pacing. You can expect about ten quizzes and three projects. The quiz format and frequency will be announced in class. You will get a prior notice before each quiz. While the above grading distribution will determine your grade, **you must pass the final to pass the class**.

8 Project Submission

Each project will have its own set of instructions that you must follow in order to get full credit. Projects will be managed via your personal, private GitHub repositories and submitted directly to Gradescope. You will receive an invitation to Gradescope on your email. If you don't receive the invitation or if you have a problem with Gradescope, email me your first and last name, emplid, and the email you want to use for Gradescope. You will have unlimited submissions to Gradescope. However, only your latest submission will count. Proper documentation and coding style will be worth 15 points for each assignment to promote good coding practices.

9 Lateness Policy

Manage your time wisely and avoid procrastination. I understand that things happen in life unexpectedly, which is why **each student gets one attempt at submitting an assignment late**. The assignment you are trying to submit late has to be an assignment that was due recently. Late submissions will be accepted with a maximum extension of seven days from the original due date.

10 Academic Integrity

Hunter College regards acts of academic dishonesty (e.g., plagiarism, cheating on examinations, obtaining unfair advantage, and falsification of records and official documents) as serious offenses against the values of intellectual honesty. The College is committed to enforcing the CUNY Policy on Academic Integrity and will pursue cases of academic dishonesty according to the Hunter College Academic Integrity Procedures." URL: <https://hunter.cuny.edu/center-for-online-learning/online-learning-resources/academic-integrity-policy/>. Any use of generative AI tools must be in line with the usage policy for specific assignments as defined in the course of the syllabus and/or communicated by the course instructor. Copying code from another person or from a generative AI system or allowing others to copy your work for submission will be treated as an act of academic dishonesty. Furthermore, making your work publicly available in forums and/or online code sharing platforms also count as a violation of academic integrity.

11 Generative AI Policy

You are allowed and encouraged to use generative AI to learn about concepts. It can be a good supplemental learning tool outside of class. However, using AI to finish your assignments will not be tolerated. Becoming reliant on generative AI can have long term consequences in your career. Code directly copied from generative AI will be counted as a violation of academic integrity, thus applying the policy stated above. However, it is okay to learn about topics covered in class by using generative AI as long as you are not copying code or asking it to do your assignments for you.

12 Anti-harassment Statement

Bullying, cyberbullying, online hate, intimidation, threats, harassment, and pressure to share schoolwork are all forms of violence. CUNY holds a zero tolerance stance towards all such acts. The University is committed to prevention of any form of bullying, will respond promptly to threats and/or acts, and will protect victims of bullying from retaliation. As a criminal matter, the New York Attorney General defines cyberbullying as the use of email, websites, instant messaging, chat rooms, text messaging and digital cameras to antagonize and intimidate others. Disrupting a teleconferencing platform (such as Zoom/Skype/Blackboard Collaborate Ultra) is a federal crime.

13 ADA Compliance

In compliance with the ADA and with Section 504 of the Rehabilitation Act, Hunter College is committed to ensuring educational access and accommodations for all its registered students. Hunter College's students with disabilities and medical conditions are encouraged to register with the Office of AccessABILITY for assistance and accommodation. For information and appointment contact the Office of AccessABILITY located in Room E1214 or call (212) 772-4857 /or VRS (646) 755-3129.

14 CUNY Policy on Sexual-Misconduct

In compliance with the CUNY Policy on Sexual Misconduct, Hunter College reaffirms the prohibition of any sexual misconduct, which includes sexual violence, sexual harassment, and gender-based harassment retaliation against students, employees, or visitors, as well as certain intimate relationships. Students who have experienced any form of sexual violence on or off campus (including CUNY-sponsored trips and events) are entitled to the rights outlined in the Bill of Rights for Hunter College.

1. Sexual Violence: Students are strongly encouraged to immediately report the incident by calling 911, contacting NYPD Special Victims Division Hotline (646-610-7272) or their local police precinct, or contacting the College's Public Safety Office (212-772-4444).
2. All Other Forms of Sexual Misconduct: Students are also encouraged to contact the College's Title IX Campus Coordinator, Dean John Rose (jtrose@hunter.cuny.edu or 212-650-3262) or Colleen Barry (colleen.barry@hunter.cuny.edu or 212-772-4534) and seek complimentary services through the Counseling and Wellness Services Office, Hunter East 1123.

CUNY Policy on Sexual Misconduct Link: <http://www.cuny.edu/about/administration/offices/la/Policy-on-SexualMisconduct-12-1-14-with-links.pdf>.

15 Update and Change Policy

Except for changes that substantially affect implementation of the evaluation (grading) statement, this syllabus is a guide for the course and is subject to change with advance notice. Any changes to the syllabus will be announced in class and/or via Brightspace.

16 Acknowledgements

The materials used in this course were adapted from materials publicly or personally shared with me. The course aims to mimic the same course taught by Professor Justin Tojeira during Spring 2025 semester.